ic.pipeline size=auto against the tuned prime-field baseline: equal at 20 bits, a pricing-band trade at 24 and 28

2026-11-07

ic.pipeline size=auto against the tuned prime-field baseline: equal at 20 bits, a pricing-band trade at 24 and 28

Result (measured, counted GAE, local L0 host). This compares `prime-abscissa:size=auto` (B = round(0.5·r^(1/3))) with the current prime-field IC baseline of `research/prime_base_oracle_tuning_20261029`: N = 45 at 20 bits, and the kept incumbents N = 162 at 24 and N = 407 at 28. It used that study's holdout curves, the same MITM pair-table oracle, 48 targets × 2 rounds, an A/A control and matched `rho.negation_v3`.

bits (log2 r)	B auto	N tuned	auto / tuned, w = 0	w = 0.1	w = 1	reading
20 (19.62)	46	45	0.998 [0.987, 1.010]	0.997	0.990	equal
24 (23.51)	114	162	0.914 [0.883, 0.946]	0.955	1.153	order flips in the band
28 (27.57)	292	407	0.885 [0.860, 0.910]	0.923	1.107	order flips in the band

B auto is the value `size=auto` took on each holdout curve, read from the records' factor-base descriptions. It falls within 1–4 of the 2026-10-29 study's own selected N (45, 115, 288), so the r^(1/3) rule reproduces that hand tuning without a per-size search. The study kept its larger incumbents at 24 and 28 bits for the same band reason given below.

This is not an improvement under AGENTS.md §5a. At 24 and 28 bits the smaller base saves charged relation work but adds unpriced linear-algebra row operations and table probes, so the order reverses at 1 GAE per native operation. That is the same reading the 2026-10-29 study gave its own smaller selected N (115 and 288), and why it kept N = 162 and 407. The baseline is unchanged.

Distance to matched rho.negation_v3 (w = 0): auto 8.47× / 13.4× / 19.0×; tuned 8.49× / 14.6× / 21.4× at 20, 24 and 28 bits.

What this points to

At these sizes the IC pipeline's remaining lever is the native work of its linear algebra, not the base size. With m = 2 each relation row has at most three non-zeros, yet incremental Gauss pays row operations that grow with B. Reducing that native work, by a sparse or structured elimination, would move the w = 1 column and is the next candidate.

Run

item	value
specs	`specs/P20.json`, `specs/P24.json`, `specs/P28.json`
sessions	`ECBS1h0d37b0d4de7f`, `ECBS1h410a3ad0a7df`, `ECBS1h3525e40319e5`; 576 records each, all verified
audits	`audit-P*.json`: replay-all, 384/384 each, 0 problems

item	value
band tables	`band-P*.md`, `band-P*.json`
isolation	L0 container